

Field and Service Robotics — Final Technical Report

**Modeling and Control of the RoTino Wheeled Bipod:
ZMP-Based PID and MPC + TV-LQR**

Euler–Lagrange Dynamics, Configuration Space,
Controller Design and Comparison Framework

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1 Introduction

Wheeled biped robots combine the mobility and energy efficiency of wheeled platforms with the postural versatility of legged systems. By changing the length of their legs they move the height at which the upper-body mass is carried, which makes them a *Variable-Length Wheeled Inverted Pendulum* (VL-WIP). The control problem is hard for two structural reasons: the system is *underactuated*, since the pitch of the upper body has no actuator of its own and must be steered through the wheels, and it is *non-minimum phase*, since to accelerate forward the wheels must first move backwards under a forward-leaning body.

The platform studied here is **RoTino**, a 4.29 kg wheeled biped simulated in Gazebo Fortress with ROS 2 Humble and torque-controlled through `ros2_control` at 500 Hz. Two control laws are designed and compared on the same robot, world and actuation:

1. a **cascade PID built around the Zero Moment Point** (ZMP), in which the wheels place the ZMP and a capture-point loop decides where it should be (package `rotino_pid`);
2. the **MPC + TV-LQR + VMC** architecture of Cui et al. [1], in which a gain-scheduled LQR balances the pendulum, a constrained MPC plans the upper-body motion and a virtual model controller drives the legs (package `rotino_mpc`).

With respect to the previous version of this report, the sliding mode controller has been removed from the project, the PID has been redesigned around the ZMP, and the dynamic model is derived with the parameters of the RoTino URDF instead of those of the robot of the paper. The symbolic model follows the structure of the MATLAB script `Matlab/SMC/init.m` and is recomputed in SymPy (`docs/presentazione/modello_lagrangiano.py`), which also checks its properties automatically.

The report is structured as follows. Section 2 lists the modelling assumptions and parameters. Section 3 analyses the degrees of freedom and the topology of the configuration space. Section 4 derives the dynamic model. Sections 5 and 6 describe the two controllers. Section 7 sets up the comparison framework. Section 9 concludes.

2 Robot Parameters and Modelling Assumptions

2.1 Physical Parameters

All parameters are read from the URDF (`rotino_description/urdf/rotino.urdf.xacro`) by the `WBRModel` class: no value is copied by hand into the controllers.

Table 1: Physical parameters of RoTino (per side unless stated otherwise).

Symbol	Value	Origin
m_w	0.38 kg	wheel 0.32 + hub 0.06
I_w	$2.595 \cdot 10^{-3}$ kg m ²	wheel i_{yy} , including the MuJoCo armature
r, d	0.06 m, 0.294 m	wheel radius, track
m_b	3.53 kg	total mass (4.29 kg) minus the two wheels
m_3	2.85 kg	torso 2.50 + ballast 0.35
m_2, m_1	0.18 kg, 0.16 kg	thigh, shank
$l_1 = l_2$	0.130 m	$\sqrt{2} \cdot 0.0919$: segments inclined at 45°
μ	2.2	wheel-ground friction
$\tau_{w,max}, \tau_{leg,max}$	18 Nm, 60 Nm	URDF limits (both controllers use 10 Nm on the wheels)

Three features of the robot weigh on the control design:

1. **Heavy rolling wheels.** $2I_w/r^2 = 1.44$ kg: the rotational inertia of the wheels, reflected to translation, is 41% of m_b and cannot be neglected.
2. **Forward torso CoM.** The torso CoM lies at $(+27.5; -5.5)$ mm from the hip; at equilibrium the torso must therefore be tilted backwards by about 4.7° .
3. **Smaller real inertia.** The inertia of the upper body about its CoM, computed from the URDF tensors, is $I_y = 0.0174$ kg m², whereas Table 1 of [1] assumes $m_b l^2/3 = 0.0311$ kg m² (1.8 times larger).

2.2 Modelling Assumptions

- **Sagittal plane.** The two legs move together and are lumped: leg and wheel masses and inertias count twice, the torso once.
- **Rigid bodies, ideal joints.** Shank and thigh are rods with the CoM at mid-length; the torso CoM is at (a_3, b_3) in the torso frame.
- **Pure rolling on flat ground.** $s_w = r q_w$, no slip, no lift-off.
- **Torque actuation.** Wheels, knees and hips receive direct torques (**effort** command interface).

Neglected effects: yaw and roll (treated separately), wheel slip and contact loss, joint damping (0.8 Nm s/rad, compensated in feedforward by both controllers), motor dynamics and sampling, camber and width of the cylindrical wheel (56 mm).

3 Topology Analysis and Degrees of Freedom

3.1 DOF Count

Before imposing the ground contact, the robot is a floating base with six joints:

$$n = \underbrace{6}_{\text{torso}} + \underbrace{2}_{\text{hips}} + \underbrace{2}_{\text{knees}} + \underbrace{2}_{\text{wheels}} = 12, \quad \mathbf{q} = [x, y, z, \phi, \theta, \psi, q_{hl}, q_{hr}, q_{kl}, q_{kr}, q_{wl}, q_{wr}]^T. \quad (1)$$

The torso is a free rigid body (three translations and three rotations); hips and knees are revolute joints limited to ± 0.9 rad; the wheels are continuous joints.

3.2 Topology of the Configuration Space

Spaces with the same number of DOFs are not necessarily equivalent topologically. For RoTino

$$\mathcal{C} = SE(3) \times \mathbb{R}^4 \times T^2 \cong \mathbb{R}^3 \times SO(3) \times [q^-, q^+]^4 \times S^1 \times S^1. \quad (2)$$

The torso lives in $SE(3)$, and $SO(3)$ is represented with quaternions to avoid the gimbal lock. Hips and knees have mechanical limits: each joint is a closed interval, not a circle, hence $\mathbb{R}^4$. The wheels rotate without limits: two circles, i.e. a torus T^2 . $\mathbb{R}^{12}$ and $\mathcal{C}$ cannot be deformed into each other.

3.3 Holonomic and Non-Holonomic Constraints

When the wheels touch the ground two kinds of constraints appear. The *holonomic* ones (flat ground: axle height fixed at r , roll tied to the legs) reduce the number of coordinates. The *non-holonomic* ones, in Pfaffian form $A^T(\mathbf{q})\dot{\mathbf{q}} = 0$, express rolling without slipping:

$$\dot{x} \sin \psi - \dot{y} \cos \psi = 0 \quad (\text{no lateral slip}), \quad (3)$$

$$\dot{x} \cos \psi + \dot{y} \sin \psi = \frac{r}{2}(\omega_l + \omega_r), \quad \dot{\psi} = \frac{r}{d}(\omega_r - \omega_l). \quad (4)$$

They restrict the admissible velocities but not the reachable configurations, since the Lie brackets of the input vector fields span the tangent space. The constraint reactions enter the dynamics through Lagrange multipliers:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{B}(\mathbf{q})\boldsymbol{\tau} + \mathbf{A}(\mathbf{q})\boldsymbol{\lambda}. \quad (5)$$

Brockett's theorem [4]. The differential-drive (unicycle) model admits no smooth, time-invariant feedback that asymptotically stabilises a point (x_d, y_d, ψ_d) . Both controllers therefore stabilise the balance and *track trajectories in time* rather than parking on a pose.

3.4 Underactuation at Three Levels

Underactuation is a property of the model in use. Table 2 lists the three models that appear in this report.

Table 2: Coordinates and actuators of the three models.

Model	n	m	Remark
Free robot	12	6	wheels, knees, hips; rank $\mathbf{B} = 6 < n$
Planar model (Sec. 4)	4	3	q_1 has no motor; the motor reactions move it (non-collocated)
Sagittal VL-WIP	2	1	s and θ with the single input $u = \tau_l + \tau_r$

The consequence for control is that *to move forward the robot must first lean*: the lean angle, or equivalently the ZMP, is the virtual input of the translation. The two controllers differ in how they choose it: the PID through the ZMP, the MPC through the CoM offset Δs .

4 Dynamic Model

4.1 Generalised Coordinates

The planar model uses the four coordinates of `init.m` (Fig. 1):

$$\mathbf{q} = [q_1, q_w, q_2, q_3]^T \quad (6)$$

where q_1 is the absolute shank angle from the horizontal (passive), q_w the absolute wheel rotation ($s_w = r q_w$, actuated by τ_w), q_2 the relative knee angle (thigh = $q_1 + q_2$, actuated by τ_k) and q_3 the absolute torso pitch from the vertical (actuated by the hip torque τ_h). In the nominal pose of RoTino (all URDF joints at zero) $q_1 = 135^\circ$, $q_2 = -90^\circ$, $q_3 = 0$: the shank rises backwards and the thigh comes forward, so that the wheel lies exactly under the hip. The mapping from the URDF joints is $q_1 = 135^\circ - (\theta_{pitch} + q_{hip} + q_{knee})$, $q_2 = q_{knee} - 90^\circ$, $q_3 = \theta_{pitch}$.

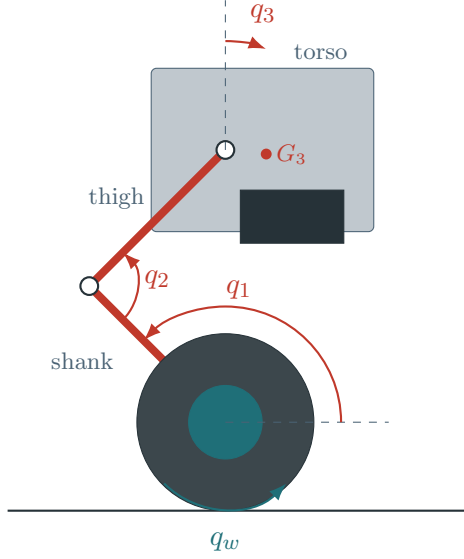


Figure 1: Planar model of RoTino in the nominal pose (to scale): generalised coordinates and torso CoM G_3 . q_1 and q_2 are counter-clockwise, q_3 and q_w clockwise (forward), as in `init.m`.

4.2 CoM Kinematics and Energies

Starting from the wheel centre, the CoM positions are written along the chain:

$$\mathbf{p}_1 = \begin{bmatrix} s_w \\ 0 \end{bmatrix} + \frac{l_1}{2} \begin{bmatrix} \cos q_1 \\ \sin q_1 \end{bmatrix}, \quad \mathbf{p}_k = \begin{bmatrix} s_w \\ 0 \end{bmatrix} + l_1 \begin{bmatrix} \cos q_1 \\ \sin q_1 \end{bmatrix}, \quad (7)$$

$$\mathbf{p}_2 = \mathbf{p}_k + \frac{l_2}{2} \begin{bmatrix} \cos q_{12} \\ \sin q_{12} \end{bmatrix}, \quad \mathbf{p}_h = \mathbf{p}_k + l_2 \begin{bmatrix} \cos q_{12} \\ \sin q_{12} \end{bmatrix}, \quad (8)$$

$$\mathbf{p}_3 = \mathbf{p}_h + a_3 \begin{bmatrix} \cos q_3 \\ -\sin q_3 \end{bmatrix} + b_3 \begin{bmatrix} \sin q_3 \\ \cos q_3 \end{bmatrix}, \quad q_{12} = q_1 + q_2. \quad (9)$$

The only difference with `init.m` is the last line: the torso CoM is shifted by $(a_3, b_3) = (+27.5; -5.5)$ mm in the torso frame instead of lying at $l_3/2$ above the hip. The kinetic and potential energies are

$$T = \frac{1}{2}(m_w r^2 + I_w) \dot{q}_w^2 + \sum_{i=1}^3 \frac{1}{2} m_i \dot{\mathbf{p}}_i^T \dot{\mathbf{p}}_i + \frac{1}{2} I_1 \dot{q}_1^2 + \frac{1}{2} I_2 (\dot{q}_1 + \dot{q}_2)^2 + \frac{1}{2} I_3 \dot{q}_3^2, \quad (10)$$

$$U = m_w g r + \sum_{i=1}^3 m_i g (r + y_i), \quad \mathcal{L} = T - U, \quad (11)$$

with $\dot{\mathbf{p}}_i = (\partial \mathbf{p}_i / \partial \mathbf{q}) \dot{\mathbf{q}}$. With the legs lumped, the parameters are $m_1 = 0.32$, $m_2 = 0.36$, $m_3 = 2.85$ and $m_w = 0.76$ kg, $I_1 = 4.62 \cdot 10^{-4}$ and $I_2 = 5.25 \cdot 10^{-4}$ kg m², $I_3 = 8.13 \cdot 10^{-3}$ kg m² and $I_w = 5.19 \cdot 10^{-3}$ kg m².

4.3 Euler–Lagrange Equations

The Euler–Lagrange equations give the standard structured form [7]

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{B} \boldsymbol{\tau} \quad \implies \quad \mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{B} \boldsymbol{\tau} \quad (12)$$

with

$$\mathbf{M}(\mathbf{q}) = \frac{\partial^2 T}{\partial \dot{\mathbf{q}} \partial \dot{\mathbf{q}}}, \quad \mathbf{G}(\mathbf{q}) = \frac{\partial U^T}{\partial \mathbf{q}}, \quad c_{kj} = \sum_i \Gamma_{ijk} \dot{q}_i, \quad \Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial M_{kj}}{\partial q_i} + \frac{\partial M_{ki}}{\partial q_j} - \frac{\partial M_{ij}}{\partial q_k} \right). \quad (13)$$

Building $\mathbf{C}$ with the Christoffel symbols of the first kind guarantees the skew-symmetry of $\dot{\mathbf{M}} - 2\mathbf{C}$.

4.4 Mass Matrix

Collecting the coefficients

$$k_1 = \left(\frac{m_1}{2} + m_2 + m_3\right)l_1, \quad k_2 = \left(\frac{m_2}{2} + m_3\right)l_2, \quad J_1 = I_1 + \left(\frac{m_1}{4} + m_2 + m_3\right)l_1^2, \quad J_2 = I_2 + \left(\frac{m_2}{4} + m_3\right)l_2^2, \quad (14)$$

and the projection of the torso CoM $\psi(\alpha) = a_3 \cos \alpha + b_3 \sin \alpha$, $\psi'(\alpha) = -a_3 \sin \alpha + b_3 \cos \alpha$, with the shorthand $s_{12} = \sin(q_1 + q_2)$, $\psi_{13} = \psi(q_1 + q_3)$, $\psi_{123} = \psi(q_1 + q_2 + q_3)$, the mass matrix reads

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} J_1 + J_2 + 2l_1 k_2 \cos q_2 & -r(k_1 \sin q_1 + k_2 s_{12}) & J_2 + l_1 k_2 \cos q_2 & -m_3(l_1 \psi_{13} + l_2 \psi_{123}) \\ \star & I_w + (m_w + m_1 + m_2 + m_3)r^2 & -r k_2 s_{12} & m_3 r \psi'(q_3) \\ \star & \star & J_2 & -m_3 l_2 \psi_{123} \\ \star & \star & \star & I_3 + m_3(a_3^2 + b_3^2) \end{bmatrix}. \quad (15)$$

For RoTino $k_1 = 0.438$ kg m and $k_2 = 0.394$ kg m. $M_{22} = 0.0206$ kg m² (translating mass plus wheel inertia) and $M_{44} = 0.0104$ kg m² (rigid torso about the hip) are constant. The terms M_{12} , M_{23} , M_{24} couple the forward motion with the posture of legs and torso: they are the root of the non-minimum-phase behaviour. The compact form is checked in the script against the expanded SymPy expression.

4.5 Coriolis, Centrifugal and Gravity Terms

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} = \begin{bmatrix} -l_1 k_2 \sin q_2 (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) - m_3 \dot{q}_3^2 (l_1 \psi'_{13} + l_2 \psi'_{123}) \\ -r(k_1 \cos q_1 \dot{q}_1^2 + k_2 \cos(q_1 + q_2)(\dot{q}_1 + \dot{q}_2)^2 + m_3 \psi(q_3) \dot{q}_3^2) \\ l_1 k_2 \sin q_2 \dot{q}_1^2 - m_3 l_2 \psi'_{123} \dot{q}_3^2 \\ -m_3(l_1 \psi'_{13} \dot{q}_1^2 + l_2 \psi'_{123} (\dot{q}_1 + \dot{q}_2)^2) \end{bmatrix}, \quad \mathbf{G}(\mathbf{q}) = g \begin{bmatrix} k_1 \cos q_1 + k_2 \cos(q_1 + q_2) \\ 0 \\ k_2 \cos(q_1 + q_2) \\ -m_3 \psi(q_3) \end{bmatrix}. \quad (16)$$

Only quadratic velocity terms appear, and none in $\dot{q}_w$; neither $\mathbf{M}$ nor $\mathbf{G}$ depends on q_w , which is a *cyclic* coordinate (the ground is flat and homogeneous). $G_w = 0$: on flat ground the wheel does not feel gravity.

4.6 Numerical Values and Properties

In the nominal pose $\mathbf{q}_0$:

$$\mathbf{M}(\mathbf{q}_0) = \begin{bmatrix} 0.1062 & -0.0353 & 0.0502 & 0.0029 \\ -0.0353 & 0.0206 & -0.0167 & -0.0009 \\ 0.0502 & -0.0167 & 0.0502 & -0.0057 \\ 0.0029 & -0.0009 & -0.0057 & 0.0104 \end{bmatrix} \text{ kg m}^2, \quad \mathbf{G}(\mathbf{q}_0) = \begin{bmatrix} -0.307 \\ 0 \\ 2.732 \\ -0.768 \end{bmatrix} \text{ Nm}. \quad (17)$$

The script `modello_lagrangiano.py` checks automatically that the model satisfies the following properties:

- $\mathbf{M} = \mathbf{M}^T \succ 0$, with eigenvalues $\{0.0070, 0.0082, 0.0245, 0.1476\}$ and a condition number of about 21;
- $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew-symmetric, hence $\dot{\mathbf{q}}^T(\dot{\mathbf{M}} - 2\mathbf{C})\dot{\mathbf{q}} = 0$ (passivity);
- the total mass is 4.29 kg;

- the upper-body CoM relative to the axle, $S_C = +13.3$ mm and $Z_C = 162.2$ mm, coincides with the one computed by `WBRModel.equivalent_centroid`, the library used by the controllers. The CoM is ahead of the axle with the torso level: to balance, the robot must rotate the torso backwards.

4.7 Input Matrix and Rigid-Motion Check

Each motor exerts equal and opposite torques on two bodies, so its column of $\mathbf{B}$ is the gradient of the relative angle between those bodies, with both angles measured in the same sense:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\tau} = \begin{bmatrix} \tau_w \\ \tau_k \\ \tau_h \end{bmatrix}, \quad \mathbf{B}^T = \frac{\partial}{\partial \mathbf{q}} \begin{bmatrix} \varphi_w \\ \varphi_k \\ \varphi_h \end{bmatrix} = \frac{\partial}{\partial \mathbf{q}} \begin{bmatrix} q_w + q_1 \\ q_2 \\ q_3 + q_1 + q_2 \end{bmatrix}. \quad (18)$$

The row of q_1 is not zero: the wheel and hip reactions push the shank, and the system is *non-collocated*.

A simple consistency check is a rigid rotation of the whole robot, $\delta \mathbf{q}_{rig} = [1, -1, 0, -1]^T \delta$: no relative angle changes, so the motors must do no work and $\mathbf{B}^T \delta \mathbf{q}_{rig} = \mathbf{0}$. The matrix above passes the test. The matrix used in `wbr_terms.m`, $[-1 \ 0 \ -1; \ 1 \ 0 \ 0; \ 0 \ 1 \ -1; \ 0 \ 0 \ 1]$, gives -2δ on the wheel and hip columns: with the conventions of `init.m` (q_3 and q_w clockwise, q_1 counter-clockwise) its motor reactions are not equal and opposite, and two signs should be revised. A second remark on the MATLAB code: the line `CdQ = jacobian(dLddq,Q)*dQ` of `init.m` omits the term $-\partial T/\partial \mathbf{q}$ of the Lagrange equation, whereas the generated `calc_dynamics.m` contains it (row 3: $+k \sin q_2 \dot{q}_1^2$), so it was produced by a different version of the script. Here $\mathbf{C}$ is built with the Christoffel symbols and verified against $\frac{d}{dt} \partial \mathcal{L} / \partial \dot{\mathbf{q}} - \partial \mathcal{L} / \partial \mathbf{q}$.

4.8 Non-Collocated Partial Feedback Linearisation

Following `wbr_terms.m`, the passive coordinate $q_u = q_1$ is separated from the active ones $\mathbf{q}_a = [q_w, q_2, q_3]$:

$$\begin{bmatrix} M_{uu} & M_{ua} \\ M_{au} & M_{aa} \end{bmatrix} \begin{bmatrix} \ddot{q}_u \\ \ddot{\mathbf{q}}_a \end{bmatrix} + \begin{bmatrix} h_u \\ \mathbf{h}_a \end{bmatrix} = \begin{bmatrix} B_u \\ \mathbf{B}_a \end{bmatrix} \boldsymbol{\tau}. \quad (19)$$

With the Schur complement

$$D = B_a - M_{au} M_{uu}^{-1} B_u, \quad \bar{M} = M_{aa} - M_{au} M_{uu}^{-1} M_{ua}, \quad \bar{h} = \mathbf{h}_a - M_{au} M_{uu}^{-1} h_u, \quad (20)$$

the law $\boldsymbol{\tau} = D^{-1}(\bar{M}\mathbf{v} + \bar{h})$ imposes $\ddot{\mathbf{q}}_a = \mathbf{v}$, and the passive coordinate follows the internal dynamics $\ddot{q}_u = M_{uu}^{-1}(B_u \boldsymbol{\tau} - M_{ua} \mathbf{v} - h_u)$, which still contains the instability of the pendulum. In the nominal pose (with the corrected $\mathbf{B}$) $\det D = 1.30$ and the diagonal of $\bar{M}$ is $\{0.009, 0.026, 0.010\}$ kg m².

4.9 Reduction to the VL-WIP

The controllers do not use the four-coordinate model but the VL-WIP of [1]. The upper body (torso, thighs, shanks) is reduced to a point of mass $m_b = 3.53$ kg at its CoM, at distance $l = \sqrt{S_C^2 + Z_C^2} = 0.163$ m from the axle and tilted by $\theta = \text{atan2}(S_C, Z_C)$. The legs set l , the wheels balance θ . With $\boldsymbol{\phi} = [s, \theta, \varphi]^T$ and l frozen:

$$\begin{bmatrix} m_b + 2m_w + \frac{2I_w}{r^2} & m_b l \cos \theta & 0 \\ m_b l \cos \theta & m_b l^2 + I_y & 0 \\ 0 & 0 & M_{33} \end{bmatrix} \begin{bmatrix} \ddot{s} \\ \ddot{\theta} \\ \ddot{\varphi} \end{bmatrix} + \begin{bmatrix} -m_b l \dot{\theta}^2 \sin \theta \\ -m_b g l \sin \theta \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{r} & \frac{1}{r} \\ -1 & -1 \\ -\frac{d}{2r} & \frac{d}{2r} \end{bmatrix} \begin{bmatrix} \tau_l \\ \tau_r \end{bmatrix}. \quad (21)$$

The yaw is decoupled ($\mathbf{M}$ block-diagonal, $C_3 = 0$), so balance and steering are controlled separately. The third row of the input matrix is $[-d/2r, d/2r]$ instead of the paper's $[d/2r, -d/2r]$ because the left wheel lies at $+y$ (REP 103): a change of convention, not of physics. The model loses the dynamics of l ($\dot{l}, \ddot{l}$), the rotation of the torso relative to the equivalent pendulum, and the compliance of the legs; this is acceptable as long as the legs move slowly compared with the balance, which is not the case during the jump thrust.

4.10 Linearisation and Structural Properties

Around $\theta = 0$, with $X = [s, \theta, \varphi, \dot{s}, \dot{\theta}, \dot{\varphi}]^T$:

$$\ddot{s} = a_1\theta + b_1(\tau_l + \tau_r), \quad \ddot{\theta} = a_2\theta + b_2(\tau_l + \tau_r), \quad \ddot{\varphi} = b_3(\tau_r - \tau_l), \quad (22)$$

$$a_1 = -\frac{g l^2 m_b^2 r^2}{\Delta}, \quad a_2 = \frac{g l m_b (2I_w + (m_b + 2m_w)r^2)}{\Delta}, \quad (23)$$

$$b_1 = \frac{r(I_y + l m_b(l + r))}{\Delta}, \quad b_2 = -\frac{2I_w + r(l m_b + (m_b + 2m_w)r)}{\Delta}, \quad (24)$$

with $\Delta = 2I_w(I_y + m_b l^2) + (2l^2 m_b m_w + I_y(m_b + 2m_w))r^2$ and $b_3 = dr/(2I_z r^2 + d^2(I_w + m_w r^2)) = 34.88 \text{ rad}/(\text{s}^2 \text{ Nm})$.

Table 3: VL-WIP coefficients along the pose grid, with the real I_y (bold: nominal pose).

l [m]	a_1 [m/s ²]	a_2 [s ⁻²]	b_1	b_2	$\sqrt{a_2}$ [rad/s]
0.2188	-12.02	89.17	8.055	-38.20	9.44
0.1627	-10.60	105.82	7.933	-50.15	10.29
0.1061	-7.86	120.35	7.379	-68.44	10.97
0.0738	-5.32	116.94	6.564	-80.40	10.81

Instability. The open-loop pitch has poles $\pm\sqrt{a_2}$: in the nominal pose the unstable pole is at $+10.29 \text{ rad/s}$, a divergence time constant of 97 ms.

Non-minimum phase. From the total torque u to the axle position

$$\frac{S(\sigma)}{U(\sigma)} = \frac{b_1\sigma^2 + (a_1b_2 - a_2b_1)}{\sigma^2(\sigma^2 - a_2)}, \quad (25)$$

with real zeros at $\pm 6.23 \text{ rad/s}$. To accelerate forward the CoM must be ahead of the axle, and to bring it there the wheels must first move backwards; no causal controller avoids this initial undershoot.

Static lean-to-acceleration gain. Holding the pendulum at a constant lean ($\ddot{\theta} = 0$) gives $\ddot{s} = (a_1 - b_1a_2/b_2)\theta = g_{st}\theta$ with $g_{st} = 6.14 \text{ m}/(\text{s}^2 \text{ rad})$: *the lean is the input of the translation*. Both controllers rest on this mechanism.

5 Cascade PID Controller with the Zero Moment Point

The controller (`rotino_pid`) comes from the port of a MuJoCo balance-and-jump controller and has been redesigned around the ZMP. The control law is in `zmp_balance.py`, pure NumPy and unit-tested offline; the ROS node is `controller.py`.

5.1 The Idea: the ZMP as Input

In the cart-table / LIPM model [2]

$$\ddot{c} = \omega^2(c - p), \quad \omega^2 = \frac{g}{h}, \quad \xi = c + \frac{\dot{c}}{\omega}, \quad \dot{\xi} = \omega(\xi - p), \quad (26)$$

where c is the CoM, p the ZMP and $h \approx 0.19$ m the CoM height ($\omega \approx 7.1$ rad/s). On two wheels the support polygon degenerates into a segment: longitudinally the ZMP coincides with the contact, and *the wheels move the contact, hence the ZMP*. The ZMP is therefore the natural control input. ξ is the capture point [3]: where the ZMP should be to bring the robot to rest.

5.2 Architecture

Figure 2 shows the four loops. Loop (A) handles balance and forward motion on the common-mode wheel torque τ_c ; loops (B) and (C) handle heading and lateral guidance on the differential torque τ_d ; loop (D) drives the legs, with a seven-state supervisor for the jump. The two wheel torques are $\tau_l = \tau_c - \tau_d$, $\tau_r = \tau_c + \tau_d$, with priority to the common mode and a 10 Nm limit per wheel.

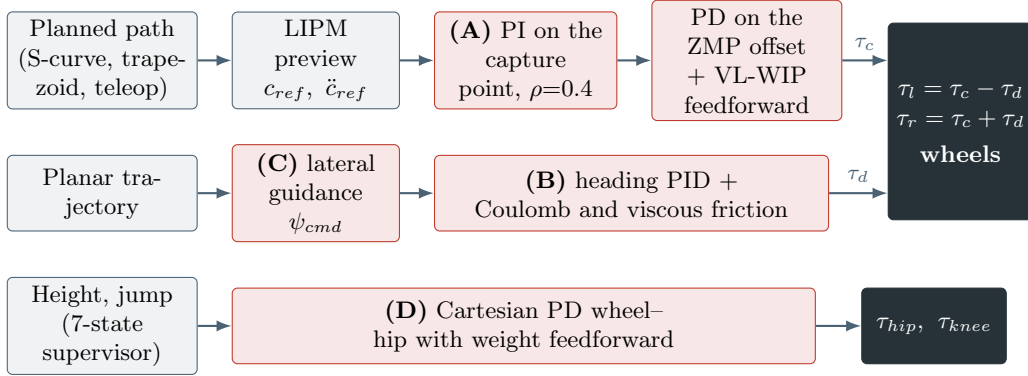


Figure 2: Loops of the ZMP-based PID. State: torso pose from `/rotino/odom` (ground truth) and URDF kinematics, paired by timestamp at 500 Hz; derivatives by finite differences.

5.3 Loop A: Capture Point $\rightarrow$ ZMP $\rightarrow$ Torque

Outer loop. A PI on the capture-point error, imposing $\dot{\xi} = \dot{\xi}_{ref} - k_{\xi}e_{\xi} - k_i \int e_{\xi}$, gives the desired offset between CoM and ZMP:

$$s_{des} = s_{ff} - \rho \left[\frac{\dot{e}_c}{\omega} + \frac{k_{\xi}}{\omega} e_{\xi} + \frac{k_i}{\omega} \int e_{\xi} dt \right], \quad |s_{des}| \leq \frac{a_{max}}{\omega^2}. \quad (27)$$

Inner loop. A PD on the offset s moves the contact where it is needed:

$$\tau_c = \tau_{ff} + K_s (s - s_{des}) + K_{sd} (\dot{s} - \dot{s}_{ff}). \quad (28)$$

Feedforward. From the linearised VL-WIP, the lean and torque that sustain the reference acceleration in steady state: $s_{ff} = \lambda_s \ddot{c}_{ref}$, $\tau_{ff} = \lambda_{\tau} \ddot{c}_{ref}$, with $\lambda_s = 21.8$ mm per m/s^2 .

Table 4: Gains of loop A.

Gain	Value
K_s, K_{sd} (per wheel)	60 Nm/m, 5 Nm s/m
ρ (authority)	0.4
k_{ξ}, k_i	$1.5 \text{ s}^{-1}, 0.5 \text{ s}^{-2}$
a_{max}	3 m/s^2 (offset ≤ 59 mm)
λ_s	21.8 mm per m/s^2
Saturation	10 Nm per wheel

5.4 LIPM Preview

An inverted pendulum cannot follow an acceleration step without leaning first. The CoM reference is the *bounded* solution of $\ddot{c} = \omega^2(c - p_{ref})$, obtained by filtering the planned contact path with a symmetric kernel that looks about $1/\omega$ into the future:

$$c_{ref}(t) = \int_{-\infty}^{+\infty} \frac{\omega}{2} e^{-\omega|u|} p_{ref}(t+u) du \quad \Rightarrow \quad \ddot{c}_{ref} = \omega^2(c_{ref} - p_{ref}). \quad (29)$$

The robot leans *before* the acceleration step, as required by the non-minimum phase. With dashboard commands the future is unknown, and the current reference acceleration, filtered at 0.1 s, is used instead. Figure 3 shows the desired and actual ZMP moving behind the CoM about 0.2 s before the start of a 1 m/s trapezoid.

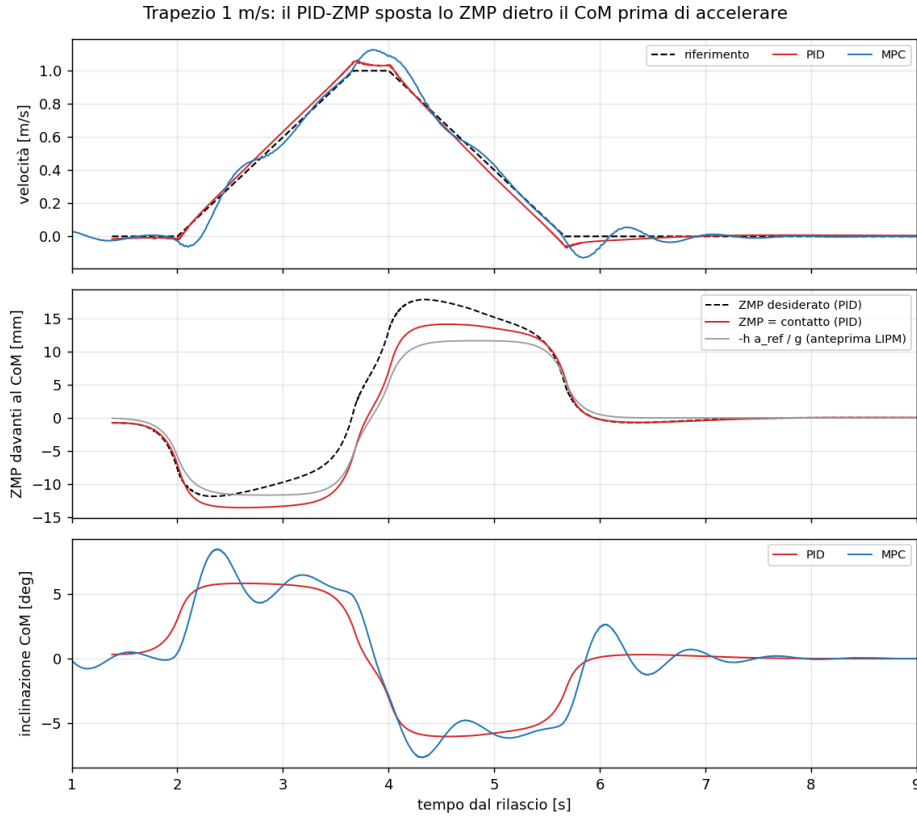


Figure 3: Trapezoidal profile at 1 m/s: velocity, desired and actual ZMP of the PID, and CoM lean. The ZMP moves behind the CoM before the wheels accelerate.

5.5 Robust Gain Design

The gains are not tuned by trial in Gazebo but chosen on the linearised model discretised at 500 Hz. The pure LIPM cascade assumes an infinitely fast inner loop; with 1–2 samples of delay and the wheel–pendulum mode at about 10 rad/s, the term $\dot{c}/\omega$ at full authority becomes positive feedback on the wheel speed and the cascade is unstable for any sufficiently high inner gain. The capture-point correction is therefore passed to the ZMP with **authority** $\rho = 0.4$; the residual error is recovered by the integral.

The gains result from robust pole placement over ten cases: mass $\pm 20\%$, hip ± 0.3 rad (CoM from crouched to stretched), delay of 1 or 2 samples. The nominal closed-loop poles are -0.5 ; $-2.3 \pm 0.2j$; -9.8 ; -52 s^{-1} , with **minimum damping 0.69** over all cases. Sixteen offline tests (`test_zmp_balance.py`) verify robust recovery from a lean, tracking of the trapezoid within

1 cm, rejection of a 2.7 N s push, removal of a CoM offset unknown to the model by the integral, and the LIPM preview.

5.6 Heading, Lateral Guidance and Legs

Heading (B) and guidance (C).

$$\tau_d = K_\psi e_\psi + K_\omega(\dot{\psi}_{ref} - \dot{\psi}) + f_c \tanh\left(\frac{\dot{\psi}_{ref}}{\varepsilon}\right) + f_v \dot{\psi}_{ref} + K_I \int e_\psi dt, \quad (30)$$

$$\psi_{cmd} = \psi_{ref} - \arctan(K_{lat} e_\perp) \min\left(1, \frac{|v_{ref}|}{0.1}\right), \quad (31)$$

with $K_\psi = 1.8$ Nm/rad, $K_\omega = 0.45$ Nm s/rad, $K_I = 0.6$, saturation 2.7 Nm and $K_{lat} = 2$ rad/m. The Coulomb (0.25 Nm) and viscous (0.10 Nm s/rad) scrub friction terms were identified in Gazebo; they act on the reference, hence without chattering. Without them the robot turned in jerks (stick-slip). The guidance correction fades out near standstill, because a differential-drive vehicle can only cancel the lateral error while moving.

Legs (D).

$$\tau_{leg} = J^T(\mathbf{q}) \left[K_p (\mathbf{p}_{des} - \mathbf{p}) - K_d J\dot{\mathbf{q}} + \mathbf{F}_{ff} \right] + b \dot{\mathbf{q}}, \quad \mathbf{F}_{ff} = -\frac{1}{2} m_b g \hat{\mathbf{z}}_b, \quad (32)$$

where $\mathbf{p}$ is the wheel centre relative to the hip (`leg_fk`). In stance $K_p = \text{diag}(1500, 5000)$ N/m and $K_d = \text{diag}(40, 80)$ N s/m; in the jump (3000, 4000); in flight (800, 800) without support. In joint space this is about 50 Nm/rad: it replaced the joint PD of the MuJoCo port (160–200 Nm/rad), which at 500 Hz entered a bang-bang limit cycle.

5.7 Jump Supervisor

The jump is a seven-state machine (BALANCE $\rightarrow$ PRELOAD $\rightarrow$ THRUST $\rightarrow$ FLIGHT $\rightarrow$ LANDING $\rightarrow$ RECOVERY $\rightarrow$ SETTLE $\rightarrow$ BALANCE). The joint references come from the MuJoCo controller and are converted into Cartesian wheel positions with `leg_fk`.

Table 5: Phases of the jump supervisor.

State	Reference (hip, knee) [rad]	Exit
PRELOAD	(0, 0) $\rightarrow$ (+0.12, −0.22) in 0.80 s, smoothstep	after 0.80 s
THRUST	$\rightarrow$ (−0.27, +0.44) in 0.02 s, $1 - (1 - \tau)^3$	take-off, or LANDING after 0.16 s
FLIGHT	pose frozen at take-off, wheels damped	confirmed contact
LANDING	$\rightarrow$ (+0.08, −0.20) in 0.80 s, compliant legs	after 0.80 s
RECOVERY / SETTLE	back to (0, 0) in 2.0 s, then settling	0.30 s at rest, or 12 s

Take-off is confirmed only when four conditions hold together: contact lost (hysteresis, 4 ms), force below threshold, wheel lifted from the ground, and vertical CoM velocity above 0.08 m/s. During the jump phases the capture-point correction is scaled per phase and the integral is frozen.

5.8 Assessment

Strengths. With known references the LIPM preview makes the robot lean in advance and track without overshoot; the integral on the capture point removes CoM offsets the model does

not know; the law is pure, separated from the ROS node and tested offline on robustness cases; the gains come from the model (robust poles), not from trial and error in simulation.

Limits and suggestions.

- The state comes from the simulator ground truth, which is not available on a real robot and makes the comparison with the MPC asymmetric.
- $\rho = 0.4$ is fixed: little authority against pushes. A ρ scheduled during transients would help.
- Constraints appear only as saturation (a_{max}); the anti-windup does not see the reduced limit $10 - |\tau_d|$ in turns.
- The telemetry publishes $z_{ref} = z$ on `/rotino/wbr_state`, so the height error logged by the benchmark is not meaningful for the PID.
- The lateral compensation is off: the cylindrical wheel jumps from one edge to the other at the inflection of the S-curve; a crowned wheel collision in the URDF would be needed.

6 MPC + TV-LQR + VMC Controller

The second controller (`rotino_mpc`, files `controller.py` and `solvers.py`) implements the architecture of Fig. 4 of [1].

6.1 Why This Architecture

The paper positions itself between the pure wheeled inverted pendulum (simple, but ignoring the torso) and whole-body control (complete, but expensive). It decouples the robot into two simple models and gives each one the most suitable controller:

- **Balance: fast and unstable.** A linear model with varying parameters (l): an LQR recomputed in l , which at run time is a matrix–vector product.
- **Posture: slower and constrained.** Where to put the CoM and how much force to ask from the legs, under friction, workspace and F_z limits: an MPC.
- **Legs: virtual model.** A virtual spring–damper at the foot plus a vertical force: no inverse dynamics in stance.

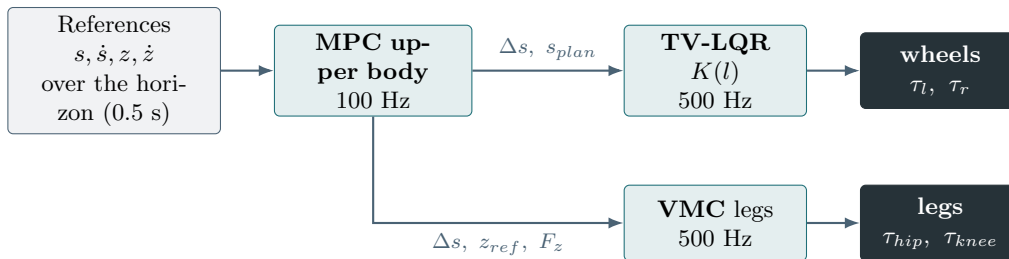


Figure 4: Architecture of the MPC + TV-LQR + VMC controller. All three blocks use the state estimated by the Kalman filter from IMU, encoders and leg kinematics.

6.2 TV-LQR Balance Loop

The cost and the gain are

$$J = \int_0^\infty (\tilde{X}^T Q \tilde{X} + U^T R U) dt, \quad U = -K(l) \tilde{X}, \quad (33)$$

$$A^T P + P A - P B R^{-1} B^T P + Q = 0, \quad K = R^{-1} B^T P, \quad (34)$$

with the Riccati equation solved as the stable subspace of the Hamiltonian. The weights are

$$Q = \text{diag}(30, 400, 80, 15, 6, 2), \quad (35)$$

$$R = T^T \text{diag}(2, 100) T, \quad T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \Rightarrow U^T R U = 2\tau_c^2 + 100 \tau_d^2. \quad (36)$$

Table 6: Bryson reading of Q : $1/\sqrt{q_{ii}}$ is the deviation that costs one unit.

State	s	θ	φ	$\dot{s}$	$\dot{\theta}$	$\dot{\varphi}$
q_{ii}	30	400	80	15	6	2
$1/\sqrt{q_{ii}}$	0.18 m	0.050 rad (2.9°)	0.11 rad	0.26 m/s	0.41 rad/s	0.71 rad/s

The lean has priority and the position is accepted within tens of centimetres: the robot yields in translation so as not to fall. The weight on the common mode coincides with $R = I$; with this R the Riccati problem splits into a sagittal and a yaw part. The differential mode costs 50 times more because of a torsional resonance of the stance legs at about 76 rad/s (torso yaw inertia on the VMC springs): with $R_d = 100$ the yaw loop crosses over at 15.4 rad/s, a fifth of the resonance.

6.3 Gains, Poles and Scheduling on l

In the nominal pose

$$K(l_0) = \begin{bmatrix} -3.87 & -20.28 & -0.89 & -5.64 & -2.78 & -0.21 \\ -3.87 & -20.28 & +0.89 & -5.64 & -2.78 & +0.21 \end{bmatrix}. \quad (37)$$

$K_s = 3.873 = \sqrt{30/2} = \sqrt{q_s/r_c}$ does not depend on l ; likewise the yaw gain $0.894 = \sqrt{q_\varphi/r_d}$. `LQRSchedule` computes K on 25 poses (each with its own I_y) and interpolates linearly in l ; the sagittal poles hardly move (Table 7), which is the purpose of the schedule. The fast pole at -180 rad/s lies beyond the bandwidth of the $\dot{\theta}$ filter, so in the real loop it is set by the filter and the sampling. The slow pair $-1.08 \pm 0.68j$ ($\zeta = 0.85$) is the return to position seen after a push. The interpolation at frozen l is justified as long as l varies slowly, which is not the case during the jump thrust, where $\dot{l}$ is ignored.

Table 7: TV-LQR gains and sagittal poles along the pose grid.

l [m]	K_θ [Nm/rad]	$K_{\dot{\theta}}$	K_s	Sagittal poles
0.219	21.22	3.13	5.51	$-139.9; -8.06; -1.15 \pm 0.68j$
0.163	20.28	2.78	5.64	$-179.5; -8.13; -1.08 \pm 0.68j$
0.074	18.43	2.33	6.32	$-281.1; -8.16; -0.86 \pm 0.63j$

6.4 Upper-Body MPC

The MPC predicts the CoM with two independent double integrators, obtained by freezing the coefficient $(g + \ddot{z})/h$ over the horizon (eq. 9 of [1]):

$$\begin{bmatrix} s \\ \dot{s} \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & \Delta T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s \\ \dot{s} \end{bmatrix}_k + \begin{bmatrix} \Delta T^2/2 \\ \Delta T \end{bmatrix} \frac{g + \ddot{z}_{ref}}{h} \Delta s_k, \quad (38)$$

$$\begin{bmatrix} z \\ \dot{z} \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & \Delta T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix}_k + \begin{bmatrix} \Delta T^2/2 \\ \Delta T \end{bmatrix} \left(\frac{F_{z,k}}{m_b} - g \right). \quad (39)$$

With $\Delta T = 0.02$ s and $N = 25$ (0.5 s horizon, updated at 100 Hz), the condensed QP

$$\min_U \frac{1}{2} U^T H U + g^T U \quad \text{s.t.} \quad \mathbf{l} \leq U \leq \mathbf{h}, \quad H = \Gamma^T \bar{S} \Gamma + W I, \quad g = \Gamma^T \bar{S} (\Phi x_0 + C - x_{ref}) \quad (40)$$

(see [6]) has only box constraints, so a projected accelerated gradient is enough: FISTA [5], 60 iterations, step $1/\lambda_{max}(H)$, warm-started from the previous solution shifted by one node. Compared with the paper, the discretisation is exact (ZOH) instead of Euler, and the QPs are two and decoupled.

Weights. Horizontal: $S_h = \text{diag}(50, 20)$ on $(s, \dot{s})$, $W_h = 200$ on Δs , so that 1 cm of Δs costs as much as 2 cm of position error. Vertical: $S_v = \text{diag}(5000, 150)$, $W_v = 2 \cdot 10^{-3}$: a stiff height, because the VMC has no vertical spring in stance.

6.5 Constraints and Reference for the LQR

$$|\Delta s| \leq \min\left(\mu z, \sqrt{L_{max}^2 - z_b^2}, 0.03\right), \quad 0.3 m_b g \leq F_z \leq \{3; 6\} m_b g, \quad (41)$$

$$\theta_{ref} = \text{atan2}(\Delta s, z_{ref}). \quad (42)$$

In the nominal pose $\mu z = 0.357$ m and $\sqrt{L_{max}^2 - z_b^2} = 0.184$ m: **the active constraint is always the 3 cm cap**, added in the implementation, which limits θ_{ref} to 10.48° and the steady deceleration to $g_{st} \cdot 0.03/0.162 = 1.13$ m/s². The friction limit of the paper would allow $\text{arctan}(2.2) = 65^\circ$ of lean. $F_{min} = 0.3 m_b g$ keeps the contact in the plan; the upper bound rises from 3 to 6 times the weight during the jump thrust.

The LQR does not track the raw reference but the MPC plan:

$$X_{ref} = [s_{plan} - \Delta s, \text{atan2}(\Delta s, z_{ref}), \varphi_{ref}, \dot{s}_{plan}, 0, \dot{\varphi}_{ref}]^T. \quad (43)$$

Δs (filtered at 19 ms) is used twice: as θ_{ref} and axle set-back for the LQR, and as the foot position relative to the CoM for the VMC. Wheels and legs thus realise the same displacement.

6.6 VMC, State Estimation and Yaw

VMC. For each leg

$$\tau_{leg} = J^T [K_p(\mathbf{p}_d - \mathbf{p}_f) + K_d(\mathbf{v}_d - \mathbf{v}_f) + \mathbf{F}_{ff}] + b_{leg} \dot{\mathbf{q}}, \quad \mathbf{F}_{ff} = -\frac{1}{2} F_z \hat{\mathbf{z}}_b, \quad (44)$$

with $K_p = \text{diag}(1500, 0)$ N/m and $K_d = \text{diag}(40, 15)$ Ns/m in stance. $K_{p,z} = 0$: the height is regulated by the MPC through F_z , and a spring in parallel would be a second regulator on the same quantity. Horizontally the foot converges in 27 ms, much faster than the CoM.

Kalman filter. Three scalar filters (one per world axis), state $[p, v]$, driven by the IMU acceleration and corrected by wheel odometry plus leg kinematics:

$$x_{k|k-1} = F x_{k-1} + \Gamma a_{w,k}, \quad P_{k|k-1} = F P_{k-1} F^T + Q, \quad (45)$$

$$K_k = P_{k|k-1} (P_{k|k-1} + R)^{-1}, \quad x_k = x_{k|k-1} + K_k (y_k - x_{k|k-1}), \quad (46)$$

with $q_a = 0.5 \text{ (m/s}^2\text{)}^2$ and $R = \text{diag}(10^{-4}, 10^{-3})$. The fusion time constants are 45 ms (IMU) and 280 ms (odometry); the correction runs only in stance.

Yaw. The φ row of the TV-LQR plus Coulomb and viscous friction feedforward and an integral with a 0.035 rad dead zone; the differential torque is limited to 2.5 Nm and the common mode to $10 - |\tau_d|$.

6.7 Assessment

Strengths. Anticipation of the references (0.5 s horizon in a non-minimum-phase system); constraints enforced in the plan rather than saturated afterwards; a single synthesis for F_z covering stance, height changes and jump thrust; a realistic estimate (IMU + Kalman) that transfers to a real robot.

Limits and suggestions.

- No integral action in the sagittal plane: on the reduced model, a constant 3 N push leaves a 0.65 m steady-state error.
- The prediction model has no pitch: the MPC does not know that moving the CoM first requires tilting the pendulum, and leaves it entirely to the LQR.
- The 3 cm cap makes the friction constraint of the paper inactive and limits the deceleration.
- Suggestions: an integral state or a disturbance model in the QP; a lean constraint depending on h ; prediction on the full VL-WIP.

7 Comparison Framework

7.1 Methodology

The comparison changes only the controller node. `robot.launch.py` starts the simulation, the robot and the scenario, and receives from the control package only the name of its executable: same URDF, same Gazebo world, same torque actuation at 500 Hz, same scenario arguments. The campaign is regenerated with `ros2 run rotino_benchmark suite`. One difference of conditions must be kept in mind: the PID reads the ground-truth state, the MPC an estimated one.

7.2 Architectures Side by Side

Table 8: Synthesis of the two architectures.

	PID with ZMP	MPC + TV-LQR + VMC
Balance state	ground truth, unfiltered derivatives	IMU + Kalman + kinematics
Wheel law	PI on capture point $\rightarrow$ ZMP $\rightarrow$ PD	LQR scheduled on l
Use of the model	VL-WIP feedforward, LIPM preview	LQR (VL-WIP) + MPC (lumped mass)
Translation	LIPM preview + capture-point integral	MPC with preview + LQR on the plan
Lean limit	ZMP offset $\leq a_{max}/\omega^2$ (3 m/s ²)	$\Delta s \leq 3$ cm $\rightarrow 10.48^\circ$
Yaw	PID + friction	LQR ($R_d = 100$) + friction + integral
Legs	Cartesian PD, 1500/5000 N/m	VMC, $K_{p,z} = 0$, support from F_z
Integral action	capture point, yaw	yaw only
Sagittal poles	$-0.5; -2.3 \pm 0.2j; -9.8; -52$	$-8.13; -1.08 \pm 0.68j$

7.3 Benchmark Scenarios and Figures

The comparison is organised in four families of scenarios (Table 9). For each family the figures below indicate the plot to be inserted from `benchmark_runs/suite_pid_mpc/` and the metrics to discuss. In the benchmark plots the PID is drawn in red and the MPC in blue.

Table 9: Benchmark scenarios.

Family	Scenarios
Disturbances	push of 2.7 N s backwards on the torso 4 s after release; vertical jump at 4 s, then balance
Tracking	trapezoid: 2 m at 1 m/s with 0.6 m/s ² ; back and forth: 1 m in 8 s
S-curves	3 m S with 0.6 m lateral offset in 12 s, and fast version: 1 m in 5 s
Balance and height	balance at rest after release; sinusoidal height ± 3 cm with 2.2 s period

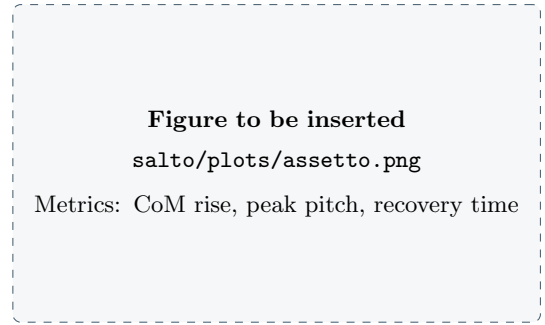
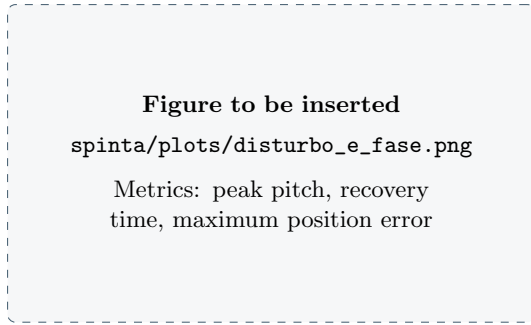


Figure 5: Disturbances: push on the torso and jump. *Key message: to be written after inserting the plots.*

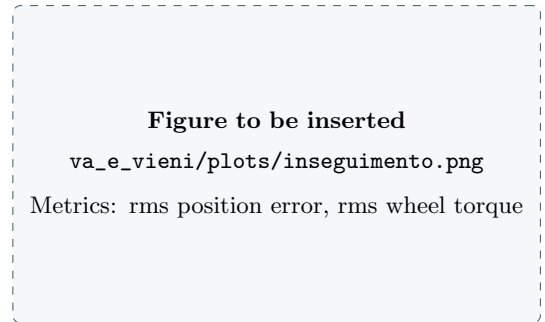
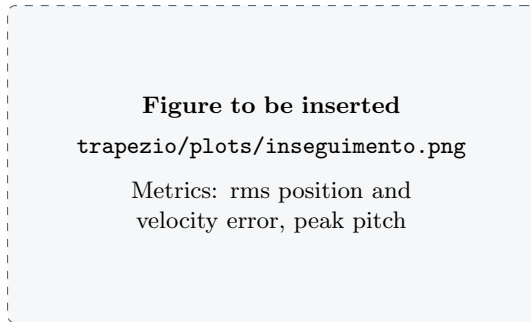


Figure 6: Tracking: trapezoid and back and forth. *Key message: to be written.*

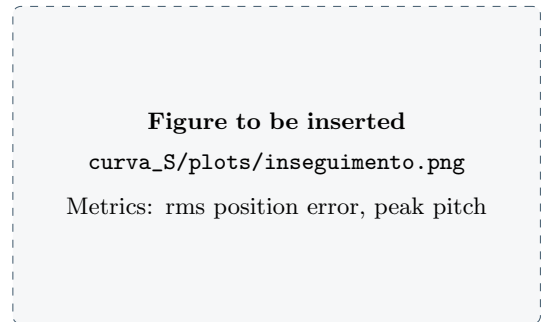
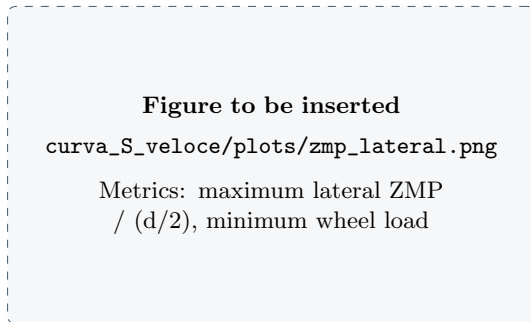


Figure 7: S-curves: tracking and lateral ZMP. *Key message: to be written.*

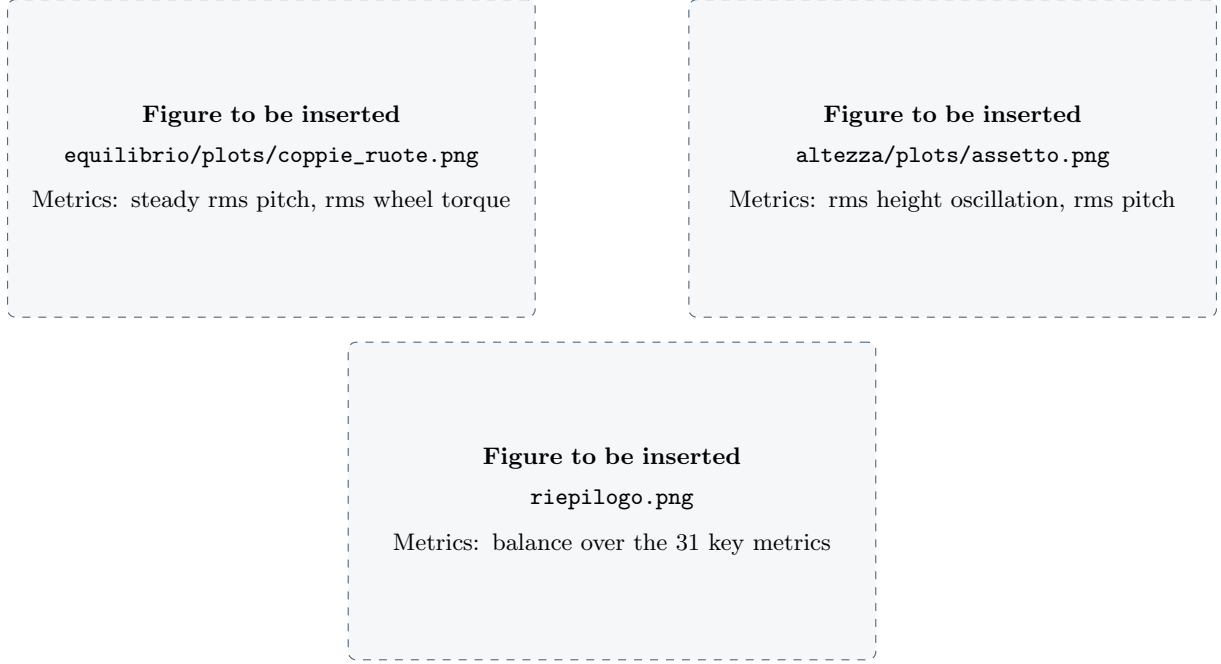


Figure 8: Balance, height change and overall balance of the metrics. *Key message: to be written.*

8 Contributions with Respect to Class Topics

- **Configuration space and DOFs:** DOF count of the floating-base robot, topology $SE(3) \times \mathbb{R}^4 \times T^2$, holonomic and Pfaffian constraints, Brockett's theorem.
- **Dynamic modelling:** symbolic Euler–Lagrange model with the parameters of the URDF, Christoffel-symbol construction of $\mathbf{C}$, automatic checks of symmetry, positive definiteness and passivity, input matrix verified with a rigid-motion test.
- **Underactuated control:** non-collocated partial feedback linearisation, reduction to the VL-WIP, analysis of instability and non-minimum phase.
- **ZMP-based balance:** cart-table model, capture point, LIPM preview, robust pole placement on the discretised model with delay.
- **Optimal control (LQR/MPC):** TV-LQR scheduled on leg length, condensed QP with FISTA at 100 Hz, common/differential torque decomposition.
- **Leg control and estimation:** Cartesian PD and virtual model control through the Jacobian transpose; linear Kalman filter fusing IMU, odometry and leg kinematics.

9 Conclusions

This report presented the modelling and control of RoTino with the parameters of its URDF, from the configuration space to two complete control laws.

1. The four-coordinate Lagrangian model, recomputed symbolically with the real parameters, reproduces the CoM of the library used by the controllers and satisfies the structural properties of a mechanical system. The analysis also exposed two inconsistencies in the MATLAB scripts (signs of $\mathbf{B}$ in `wbr_terms.m`, missing $-\partial T/\partial \mathbf{q}$ in `init.m`).
2. The reduced VL-WIP shows the two structural difficulties: an unstable pole at 10.3 rad/s and real zeros at ± 6.2 rad/s. Both controllers use the lean as the input of the translation.

3. The PID, redesigned around the ZMP, moves the contact with the wheels under a capture-point loop of reduced authority; the LIPM preview lets it lean before accelerating. Its gains come from robust pole placement on the discretised model.
4. The MPC + TV-LQR + VMC handles constraints and future references explicitly, with a realistic state estimate, but has no sagittal integral action and a prediction model without pitch.
5. The quantitative comparison is set up in Section 7 and will be completed with the plots of the benchmark suite.

Future work: a crowned wheel collision to enable the lateral ZMP compensation, a scheduled authority ρ and IMU-based estimation for the PID, an integral state and a pitch-aware prediction model for the MPC, and validation on hardware.

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